SEMESTER - VI

COURSE: MAJOR – 12 (MJ-12) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: MODERN PHYSICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course provides a comprehensive understanding of atomic and molecular spectra, quantum mechanical models of atoms, and fundamental spectroscopic techniques. Students will learn the principles of radioactivity and nuclear processes, and gain foundational knowledge of laser physics, including laser operation mechanisms and applications in modern physics and technology.

Learning Outcomes:

1. Explain atomic models, quantum numbers, and the quantum mechanical treatment of the hydrogen atom.
2. Analyze fine and hyperfine spectral structures including spin-orbit coupling, Zeeman and Stark effects.
3. Describe molecular spectra using rigid rotor and harmonic oscillator models for various molecular types.
4. Understand radioactive decay processes, nuclear stability, and energy generation through fission and fusion.
5. Explain laser operation principles, including population inversion, types of lasers, and laser applications.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Atomic Spectra (15 Lectures) : Rutherford experiment, Bohr's theory of hydrogen atom, de Broglie wavelength, hydrogen spectrum effect of nuclear mass, Ritz combination principle, merits & demerits of Bohr model. Schrödinger equation of hydrogen atom in spherical coordinates, separation of variables, angular equation, radial equation, principle (n), orbital (l), orbital magnetic quantum (m_l) numbers, & selection rules, vector model of atom, Stern-Gerlach experiment, electron spin, Bohr magnetron, spin

magnetic quantum number (m_s), total angular momentum, addition of angular momenta, spin-orbit coupling, Larmor precession; symmetric and antisymmetric wave function, LS & JJ coupling, Zeeman effect, P-B effect, Stark effect. Electron spin, Pauli's exclusion principle, symmetric and antisymmetric wave function, atomic structure and periodic table, spin – orbit coupling, total angular momentum, atomic spectra, characteristics X-Ray spectra.

UNIT-II:

Molecular Spectra (10 Lectures): Molecular bonds, H^+ molecular ion, Hydrogen molecule, types of molecules (diatomic linear, symmetric top, asymmetric top and spherical top molecules), molecule as rigid rotor and harmonic oscillator, rotational, vibrational, and electronic spectra of diatomic molecules, brief account of spectroscopy (UV-VIS-IR spectroscopy, NMR, ESR, and Raman spectroscopy), basic concept of luminescence (introduction, excitation, and emission).

Radioactivity (10 Lectures): stability of the nucleus; Law of radioactive decay; Mean life and half-life; Alpha decay; Beta decay: energy released, spectrum and Pauli's prediction of neutrino; Gamma ray emission, energy-momentum conservation: electron-positron pair creation by gamma photons in the vicinity of a nucleus. Fission and fusion: mass deficit, relativity and generation of energy; Fission: nature of fragments and emission of neutrons. Fusion and thermonuclear reactions driving stellar evolution (brief qualitative discussions).

UNIT-III:

Laser Physics (10 Lectures): Einstein's A and B coefficients. Metastable states. Spontaneous and stimulated emissions. Laser rate equations for two-level and three-level lasers. Optical Pumping and Population Inversion. Three-Level and Four-Level Lasers. Ruby Laser and He-Ne Laser. Pulsed Laser and Q-switching.

SUGGESTED READINGS:

1. Introduction to Quantum Mechanics, D.J. Griffith, 2nd Ed. 2005, Pearson Education.
2. Concept of Modern Physics, Sixth Edition, Arthur Beiser, McGraw-Hill.
3. Atomic and Molecular spectra: Laser by Raj Kumar.
4. Quantum Mechanics, Eugen Merzbacher, 2004, John Wiley and Sons, Inc.
5. A Text book of Quantum Mechanics, P.M. Mathews and K. Venkatesan, 2nd Ed., 2010, McGraw Hill
6. Quantum Mechanics, Leonard I. Schiff, 3rd Edn. 2010, Tata McGraw Hill.
7. Quantum Mechanics, G. Aruldhas, 2nd Edn. 2002, PHI Learning of India.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. To observe and measure the spectral lines of the hydrogen atom and calculate the Rydberg constant.
2. To resolve and measure the wavelength separation between the two D-lines of Sodium D_1 and D_2 lines.
3. To compare observed spectral lines (H or He) with Bohr's theoretical predictions.
4. To observe the splitting of spectral lines in the presence of a magnetic field (Zeeman effect).
5. Simulate rate equations for two- and three-level lasers and analyze dynamics.
6. Determine rotational constants from observed microwave absorption spectra of diatomic molecules.
7. Measure spatial coherence and divergence of laser beam.
8. Numerically solve the radial and angular equations and analyze quantum numbers.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VI

COURSE: MAJOR – 13(MJ-13) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: STATISTICAL MECHANICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to provide students with a thorough understanding of classical and quantum statistics, including the concepts of macrostates and microstate, ensembles, entropy, distribution laws (Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac), and their applications in describing the behavior of particles in various physical systems.

Learning Outcomes:

1. Understand the concepts of microstates, macrostates, ensembles, and entropy in classical statistical mechanics.
2. Apply Maxwell-Boltzmann distribution and partition functions to calculate thermodynamic quantities.
3. Explain blackbody radiation and derive radiation laws using Planck's quantum theory.
4. Analyze Bose-Einstein statistics, photon gas properties, and Bose-Einstein condensation phenomena.
5. Interpret Fermi-Dirac statistics and apply them to electron behavior in metals, including thermionic and photoelectric emissions.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Classical Statistics (15 Lectures): Macrostates & Microstate, Elementary Concept of Ensemble: micro-canonical, canonical, grand canonical, Phase Space, Entropy and Thermodynamic Probability, Maxwell-Boltzmann Distribution Law, Partition Function, Thermodynamic Functions of an Ideal Gas, Classical Entropy Expression, Gibbs Paradox & resolution, Sackur-Tetrode equation, Law of Equipartition of Energy (with proof), Applications to Specific Heat and its Limitations.

UNIT-II:

Quantum Theory of Radiation (10 Lectures): Spectral Distribution of Black Body Radiation, Planck's Quantum Postulates. Planck's Law of Blackbody Radiation: Experimental Verification. Deduction of (1) Wien's Distribution Law, (2) Rayleigh-Jeans Law, (3) Stefan-Boltzmann Law, (4) Wien's Displacement Law from Planck's law, Applications.

UNIT-III:

Bose-Einstein and Fermi-Dirac Statistics (20 Lectures): B-E distribution law, Thermodynamic functions of a strongly Degenerate Bose Gas, Bose Einstein condensation, properties of liquid He (qualitative description), Radiation as a photon gas and Thermodynamic functions of photon gas, Bose derivation of Planck's law. Fermi-Dirac Distribution Law, Thermodynamic functions of a strongly Degenerate Fermi Gas, Fermi Energy, Electron gas in a Metal, Specific Heat of Metals, Thermionic emission, Photoelectric Emission, Richardson equation, Pauli spin paramagnetism.

SUGGESTED READINGS:

1. Statistical Mechanics, R.K. Pathria, Butterworth Heinemann: 2nd Ed., 1996, Oxford University Press.
2. Statistical Physics, Berkeley Physics Course, F. Reif, 2008, Tata McGraw-Hill.
3. Statistical and Thermal Physics, S. Lokanathan and R.S. Gambhir, 1991, Prentice Hall.
4. Thermodynamics, Kinetic Theory and Statistical Thermodynamics, Francis W. Sears and Gerhard L. Salinger, 1986, Narosa.
5. Modern Thermodynamics with Statistical Mechanics, Carl S. Helrich, 2009, Springer.
6. An Introduction to Statistical Mechanics & Thermodynamics, R.H. Swendsen, 2012, Oxford Univ. Press.
7. An Introduction to Thermal Physics, Daniel V. Schroeder, Addison-Wesley, Reading, Massachusetts, 2000.

8. Statistical Mechanics by Satya Prakash.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Measure specific heats and verify the law of equipartition of energy and explore its limitations.
2. Visualize the distribution of microstates and macrostates using computational methods.
3. Verify the Rayleigh-Jeans approximation in the classical limit.
4. Estimate Fermi energy using thermionic emission experiments.
5. Computationally visualize occupancy of energy states in a Fermi gas at different temperatures.
6. Analyze radiation intensity distribution for different temperatures.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VI

COURSE: MAJOR – 14 (MJ-14) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: DIGITAL ELECTRONICS AND APPLICATIONS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to provide foundational knowledge of number systems, Boolean algebra, and digital logic design. It introduces combinational and sequential circuits, memory devices, and microprocessor architecture. Students will develop the skills to analyze, simplify, and implement digital systems, and gain insight into basic microprocessor operations and components.

Learning Outcomes:

1. Understand number systems, binary arithmetic, and Boolean algebra for digital logic design.
2. Simplify Boolean functions using Karnaugh maps and implement logic circuits using SOP/POS forms.
3. Design and analyze combinational and sequential circuits using basic and universal logic gates.
4. Explain the operation of flip-flops, counters, shift registers, and memory devices.
5. Understand microprocessor architecture and describe the basic components and features of 4-bit to 32-bit systems.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Number systems and Boolean algebra (20 Lectures): Positional number system, binary representation, 2's complement notation, binary addition and subtraction, octal number system, hexadecimal system, binary codes (BCD and ASCII codes), Switching circuits, AND, OR and NOT operations, truth table, Boolean functions, postulates and theorems of Boolean algebra, duality principle, Venn diagram, canonical

forms of Boolean functions, simplification of Boolean functions, sum of product (SOP) and product of sum (POS) representations, Karnaugh maps, incompletely specified functions, don't cares.

UNIT-II:

Combinational and sequential circuits (10 Lectures): Logic gates, NAND and NOR as universal gates, realization of logical functions using SOP and POS techniques, XOR gate, decoders, encoders, multiplexers, de- multiplexers, code conversion using logic gates and MSI ICs, half adder, full adder, serial adder, half subtractor, full subtractor, digital comparator, TTL & CMOS.

UNIT-III:

Registers and Memory Devices (15 Lectures): Flip-flops (J-K, S-R, clocked, master-slave, & edge-triggered flip-flops), synchronizer, timer, counter, shift register. Read only memories, shift-register memories, Random-Access Memories. Microprocessor Architecture: Introduction, 4-Bit, 8-Bit, 16-Bit, & 32-Bit systems, components, brief description of 8085/8086 microprocessor.

SUGGESTED READINGS:

1. Digital electronics by G. K. Kharate, Oxford University Press, 2010.
2. Principles of electronics, V. K. Mehta and Rohit Mehta, S. Chand.
3. Electronic Device and Circuit Theory, XIth Edition, Robert L. Boylestad and Louis Nashelsky.
4. Digital Principles and Applications, A.P. Malvino, D.P.Leach and Saha, 7th Ed., 2011, Tata McGraw
5. Fundamentals of Digital Circuits, Anand Kumar, 2nd Edn, 2009, PHI Learning Pvt. Ltd.
6. Digital Circuits and systems, Venugopal, 2011, Tata McGraw Hill.
7. Digital Electronics, SubrataGhoshal, 2012, Cengage Learning.
8. Microprocessor Architecture Programming & applications with 8085, 2002, R.S. Goankar, Prentice Hall.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Construction of half and full adder using XOR and NAND gates and verification of its operation.
2. Verify the truth table of RS, JK, T and D flip-flops using NAND and NOR gates.
3. Encode and decode characters using BCD and ASCII formats.

4. Write a program to convert numbers between binary, octal, decimal, and hexadecimal systems.
5. Design and demonstrate data shifting operations in a shift register.
6. Implement Boolean expressions using logic gates in SOP and POS formats.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VI

COURSE: MAJOR – 15 (MJ-15) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: SOLID STATE PHYSICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to provide students with a comprehensive understanding of the crystal structure of materials, including lattice translation vectors, unit cells, lattice dynamics, magnetic properties, dielectric properties, ferroelectric properties, elementary band theory, and superconductivity, enabling them to analyze the physical properties and behavior of various materials.

Learning Outcomes:

1. Understand crystal structures, lattice vectors, Miller indices, and reciprocal lattices including Brillouin zones and X-ray diffraction principles.
2. Explain lattice vibrations, phonon types, and theories of specific heat in solids like Einstein and Debye models.
3. Describe free electron theory, Fermi energy concepts, band theory, and classify solids based on band structure.
4. Analyze thermal, dielectric, magnetic, and optical properties of solids with relevant physical models.
5. Comprehend fundamentals of superconductivity including Meissner effect, critical parameters, and BCS theory overview.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Crystal Structure and Lattice Dynamics (15 Lectures): Solids: Amorphous and Crystalline Materials, Lattice Translation Vectors, Lattice with a Basis – Central and Non-Central Elements, Unit Cell, Miller Indices, Reciprocal Lattice, Types of Lattices, Brillouin Zones -reciprocal lattice to sc, bcc, &fcc lattices, , Diffraction of X-rays by Crystals, Bragg's Law, Atomic and Geometrical Factor. Lattice Vibrations and Phonons: Linear Mono-atomic and Diatomic Chains, Acoustic and Optical Phonons, Qualitative description of the Phonon Spectrum in Solids, Dulong and Petit's Law, Einstein and Debye theories of specific heat of solids, T3 law .

UNIT-II:

Energy band (10 Lectures): Free electron theory of metals - thermal & electrical conductivity, Drude-Lorentz theory, Somerfield's theory, Fermi energy, Fermi level, Fermi velocity, density of states, band theory of solids, classification of solids (metals, semiconductor, & insulators) on the basis of band theory, Brillouin zone and origin of band gap, Kronig-Penny model, effective mass of electron, direct & indirect band gap semiconductors.

UNIT-III:

Properties of solids (20 Lectures): Thermal properties- Lattice vibration, vibration of 1-D monoatomic & diatomic lattice, concept of phonon, phonon dispersion relation, specific heat of solids, Dulong-Petit law, Einstein & Debye's theory, density of states. Dielectric properties- Polarization, dielectric constant, polarizability, local electric field, Normal and Anomalous Dispersion, Cauchy and Sellmeir relations, Langevin-Debye equation, Complex Dielectric Constant. Magnetic properties- Dia-, Para-, Ferri- and Ferromagnetic Materials, Classical Langevin Theory of Diamagnetic and Paramagnetic Domains, Quantum Mechanical Treatment of Paramagnetism, Curie's law, Weiss's Theory of Ferromagnetism and Ferromagnetic Domains, Discussion of B-H Curve, Hysteresis and Energy Loss. Optical properties- Brief qualitative discussion of optical properties, complex refractive index, extinction coefficient. Superconductivity: Experimental Results, Critical Temperature, Critical magnetic field, Meissner effect, Type I and type II Superconductors, London's Equation and Penetration Depth, Isotope effect, Idea of BCS theory.

SUGGESTED READINGS:

1. Introduction to Solid State Physics, Charles Kittel, 8th Edition, 2004, Wiley India Pvt. Ltd.

2. Solid State Physics, S. O. Pillai.

3. Solid State Physics, R. K Puri and V. K Babbar, S. Chand & Company Ltd.

4. Elements of Solid State Physics, J.P. Srivastava, 4th Edition, 2015, Prentice-Hall of India

5. Solid State Physics, N.W. Ashcroft and N.D. Mermin, 1976, Cengage Learning

6. Elementary Solid State Physics, 1/e M. Ali Omar, 1999, Pearson India.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Determine the Hall coefficient of a semiconductor sample.
2. Determine the Coupling Coefficient of a Piezoelectric crystal.
3. Measure the Dielectric Constant of a dielectric Materials with frequency.
4. Measure diffraction angles for a known crystal and verify Bragg's Law.
5. Simulate phonon dispersion relation for monoatomic and diatomic chains.
6. Estimate the band gap energy using thermally activated current in a PN junction diode.

SUGGESTED READINGS:

1. https://phet.colorado.edu/

2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House

3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal

4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers

5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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